

ROLL-OFF CONTAINER MANUFACTURING FACILITY — INVESTOR REPORT

Phased Production Roadmap (Phase 1 → Phase 2 → Phase 3)

Date: August 2026 · **FX:** 1 USD ≈ ₺47.5 · 1 EUR ≈ ₺54.8 **Note:** All USD equivalents are calculated at 1 USD ≈ ₺47.5.
Document: Investment decision support — all figures are estimates based on August 2026 market data.

1. EXECUTIVE SUMMARY

This project establishes a manufacturing facility for **30 m³ heavy-duty Roll-Off containers** (6,000 × 2,500 × 2,200 mm, open top, hook bar, skid rails). Production is set up in a **three-phase phased model** rather than a single large build: each phase is financed from its own cash flow, and a go/no-go decision gate exists at every stage.

Metric	Value (base case)
Total investment (3 phases)	₺84.5 million (~\$1.78M)
Phase 1 investment	₺30 million (~\$632K)
Year 1 unit sales	268 units
Year 3 unit sales	650 units
Unit production cost (Phase 1)	~₺317,000 (~\$6,674)
Average selling price	₺500–545 thousand (~\$10,526–11,474) (entry → normalization)
Annual EBITDA (Y1 / Y2 / Y3)	₺50M / ₺107M / ₺161M (~\$1.05M / ~\$2.25M / ~\$3.39M)
EBITDA margin	~38% (entry) → 46% (Y3)
Simple payback	~15 months (from production start)
Cumulative free cash flow (3 years)	~₺159M (~\$3.35M) (~1.9× investment)

Market entry strategy (price): Enter at **₺500,000 (~\$10,526)** — ~13% below domestic competitors; growth comes from volume. Price normalizes to ₺520,000 (~\$10,947) in Y2 and ₺545,000 (~\$11,474) in Y3. With ~₺311,000 (~\$6,547) unit cash cost, entry margin is ~38%; scale + automation push it to ~46% by Y3.

Why this product? Roll-off containers serve the excavation, construction, municipal and industrial waste markets with steady demand. They are low-technology but high-value-added products. Local production competes with imports on short lead time, customization, warranty and "domestic product" status in public tenders. Customs protection on steel products limits the import price advantage.

2. PRODUCT — TECHNICAL SPECIFICATION

Parameter	Value
Capacity	~30 m ³
External dimensions	6,000 × 2,500 × 2,200 mm
Top	Open (hook-lift loading)
Floor plate	6 mm (5–8 mm range)
Wall plate	4 mm (3–5 mm range)
Chassis (skid) rails	2 × IPE 160 (I-beam)
Cross members	7 pcs, U 100 / box 80×80×4
Corner posts	Box 80×80×4
Intermediate posts	Box 60×60×3
Hook bar	Ø 50 mm solid steel + welded lugs
Rear doors	Optional double-leaf, hinged
Surface	Blasting + 2K epoxy/polyurethane paint
Container steel weight	~3,200 kg

Manufacturing approach: Plates and profiles are purchased **cut and bent** from an external supplier (Phase 1); welding and assembly are done in-house. Plates are positioned by overhead crane, profiles by crane assistance (robot-assisted from Phase 3), and the container is rotated during welding using positioner/rollover fixtures.

3. MACRO ASSUMPTIONS (AUGUST 2026)

Item	Value
FX	USD ₺47.5 · EUR ₺54.8
Minimum wage (net / gross)	₺28,075 (~\$591) / ₺33,030 (~\$695)
Minimum wage full employer cost	~₺41,500/month (~\$874/month) (employer SSI 17.5% + severance 8.3%)
Skilled welder/fabricator full cost	~₺68–76 thousand/month (~\$1,431–1,600/month)
Steel plate (S235, hot rolled)	~₺33,000/ton (~\$695/ton)
Steel profiles	~₺35,000/ton (~\$737/ton)
Scrap recovery price	~₺13,000/ton (~\$274/ton)
Corporate tax	25%
VAT	20%
Industrial facility rent	₺110–130/m ² /month (~\$2.3–2.7/m ² /month)

Note: Steel and FX prices fluctuate monthly; the model has been stress-tested with a sensitivity table (see Section 8).

4. PRODUCTION STRATEGY — THREE-PHASE ROADMAP

The model is not a single large build; it is **three independent phases**. Each phase finances the next from the cash flow it generates, and every phase ends with a clear **go/no-go gate**: if the condition is not met, no further investment is made and the existing operation is continued at reduced scale. The phases are aligned to years, so the investor can see at a glance what is spent in each phase and what revenue has been generated up to that point.

Phase comparison

	Phase 1	Phase 2	Phase 3
Timing	Year 1 (months 1–12)	Year 2 (months 13–24)	Year 3 (months 25–36)
Capacity	15–20 → 40 units/month	45 units/month	55–60 units/month
Staff	15 (9 direct / 6 indirect)	20 (12 / 8)	19 (10 / 9)
Unit cost	~£317,000 (~\$6,674)	~£314,000 (~\$6,611)	~£306,000 (~\$6,442)
Key change	Manual welding + external cutting	CNC plasma brought in-house	Robot welding goes live

Phase 1 — Year 1 (months 1–12): start production with low capital

- **Investment:** £30M (~\$632K) — workshop fit-out, overhead cranes (2 × 5 t), 2 welding stations (MIG/MAG), paint booth, portable cutting kit (~£250k ≈ \$5,263), fixtures/jigs, initial stock.
- **Method / what changes:** Cutting and bending are done at an **external supplier** (no capital required); manual welding and assembly in-house. Production starts at 15–20 units/month and ramps to ~36–40 by year-end. Building a customer portfolio realistically takes ~1 year, so no headcount is kept for 40 units/month at the start (4 welders, 1 salesperson).
- **Objective:** Validate demand with real sales, learn operations, build customer/hook-type database.

Financial position so far (end of Year 1):

Item	Value
Revenue	£134M (~\$2.82M) (268 units × £500,000 ≈ \$10,526)
EBITDA	£50.7M (~\$1.07M) (38%)
Cumulative investment (capex + working capital)	£33.5M (~\$705K)
Cumulative free cash flow	+£5.2M (~\$109K)

Gate 1 (month 12): If sustained sales are **≥ 30 units/month** by year-end (ramp ends ~36) → proceed to Phase 2. Otherwise: reduce capacity, make no further investment.

Phase 2 — Year 2 (months 13–24): bring CNC plasma cutting in-house

- **Investment:** +£6M (~\$126K) — CNC plasma + gas infrastructure (payback ~7.5 months).
- **What changes:** External cutting dependency **ends**; lead time shortens, unit cost drops from ~£317,000 (~\$6,674) to ~£314,000 (~\$6,611). Capacity rises to 45 units/month; welders 4 → 6, staff 20.

Financial position so far (end of Year 2):

Item	Value
Cumulative revenue	₺394M (~\$8.29M) (134 + 260)
Cumulative EBITDA	₺157.2M (~\$3.31M)
Cumulative investment	₺73.5M (~\$1.55M)
Cumulative free cash flow	+₺46.7M (~\$983K)

Gate 2 (month 24): If sustained sales are ≥ 45 units/month $\rightarrow$ proceed to Phase 3.

Phase 3 — Year 3 (months 25–36): welding robots and automation

- **Investment:** +₺48.5M (~\$1.02M) $\rightarrow$ **total ₺84.5M (~\$1.78M)** (₺30M + ₺6M + ₺48.5M)
- 6 $\times$ 6-DOF welding robots: ₺18M (~\$379K); EOAT/profile gripper + automatic torch changer (in-house design, material): ₺1M (~\$21K) $\rightarrow$ subtotal ₺19M (~\$400K)
- Positioner/rollover fixtures (chassis turner, body rollover, door positioner, synchronization): ₺9.5M (~\$200K)
- Jigs/locating fixtures: ₺4M (~\$84K)
- Commissioning, training, contingency: ₺16M (~\$337K)
- **What changes:** Robots take over most welding work $\rightarrow$ welders 6 $\rightarrow$ 2; a single robot operator + a welding engineer are added (staff 20 $\rightarrow$ 19). Unit cost falls to ~₺306,000 (~\$6,442), capacity rises to 55–60 units/month; consistent weld quality and resilience to welder shortage.

Financial position so far (end of Year 3):

Item	Value
Cumulative revenue	₺748M (~\$15.75M)
Cumulative EBITDA	₺318.4M (~\$6.70M)
Cumulative investment	₺84.5M (~\$1.78M)
Cumulative free cash flow	+₺158.9M (~\$3.35M) (~1.9$\times$ investment)

Why phased? Robots create a depreciation burden at low volume (a one-shot fully automated plant can be more expensive than manual at low sales). Each phase is financed from its own cash flow; there is no single-shot stranded-asset risk for the investor.

Investment breakdown (₺84.5M $\approx$ \$1.78M)

Item	₺M (~\$)
Building fit-out + overhead cranes (2 $\times$ 5 t)	6.5 (~\$137K)
Welding stations (2 $\times$ MIG/MAG + fume extraction) + gas infrastructure	5.0 (~\$105K)
CNC plasma (Phase 2)	6.0 (~\$126K)
6 welding robots + EOAT/torch changer (in-house design)	19.0 (~\$400K)
Positioner/rollover fixtures	9.5 (~\$200K)
Jigs / locating fixtures	4.0 (~\$84K)
Paint booth + surface preparation	3.0 (~\$63K)
Portable cutting kit	0.25 (~\$5K)

Item	£M (~\$)
Measuring/testing/QC equipment	2.0 (~\$42K)
Initial stock + working capital	13.25 (~\$279K)
Installation, training, commissioning	4.0 (~\$84K)
Contingency	12.0 (~\$253K)
TOTAL	84.5 (~\$1.78M)

5. UNIT COST STRUCTURE (PHASE 1, STEADY 40 UNITS/MONTH)

Item	£/unit (~\$/unit)
Steel plate (2,950 kg × £33,000/t)	97,350 (~\$2,049)
Profiles (900 kg × £35,000/t)	31,500 (~\$663)
Scrap recovery (−380 kg × £13,000)	−4,940 (~\$104)
External cutting + bending	18,000 (~\$379)
Welding consumables / gas	6,500 (~\$137)
Paint (2K)	8,000 (~\$168)
Hardware (hook, locks, hinges, door set, skid wear strips)	12,000 (~\$253)
Other consumables	3,500 (~\$74)
Material subtotal	171,910 (~\$3,619)
Direct labor (9 people, 40 units/month)	16,200 (~\$341)
Indirect labor (6 people)	10,800 (~\$227)
Management/technical staff	12,500 (~\$263)
Rent (3,500 m ²)	10,500 (~\$221)
Energy / gas	14,000 (~\$295)
Overhead, QC, insurance	12,500 (~\$263)
Logistics	9,000 (~\$189)
Sales & warranty reserve (~4%)	20,300 (~\$427)
Working capital financing	4,000 (~\$84)
Maintenance / repairs	6,000 (~\$126)
Rework / scrap	8,000 (~\$168)
Corporate / G&A allocation	15,000 (~\$316)
Operating expenses subtotal	138,800 (~\$2,922)
Depreciation (£30M / 10 years)	6,250 (~\$132)
TOTAL UNIT COST	≈ £317,000 (~\$6,673)

Cost evolution: Phase 1 ~£317,000 (~\$6,674) → Phase 2 ~£314,000 (~\$6,611) → Phase 3 ~£306,000 (~\$6,442) (with volume + automation). ~55% of unit cost is steel and material; the labor share is low because

cutting/bending is done externally.

6. PRICING AND REVENUE

Competitive entry price (penetration)

- **Entry price: ₺500,000/unit (~\$10,526)** — ~13% below domestic competitors (~₺575,000 ≈ \$12,105). Sits above the duty-included import; justified to customers by short lead time + warranty + customization.
- **Growth engine is volume, not price:** the competitive price starts market share and reference wins; scale lowers unit cost.
- **Normalization path:** Y2 ₺520,000 (~\$10,947) → Y3 ₺545,000 (~\$11,474) (gradual, as brand and references take hold).
- Options: rear doors, custom dimensions, corporate livery, special paint → +₺15–40 thousand/unit (~\$316–842)

Sales channels: Municipalities and public tenders ("domestic product" advantage), excavation/environmental contractors, construction companies, and co-selling with hook-lift truck dealers.

7. 3-YEAR FINANCIAL PROJECTIONS

Sales ramp (units/month)

- Year 1: months 3–4 at 15, months 5–6 at 20, months 7–9 at 30, months 10–12 at 36 → **annual ≈ 268 units**
- Year 2: average ~42 units/month → **500 units**
- Year 3: average ~54 units/month → **650 units**

Summary table

	Year 1	Year 2	Year 3
Units sold	268	500	650
Average price (₺ / ≈ \$)	500,000 (~\$10,526)	520,000 (~\$10,947)	545,000 (~\$11,474)
Revenue (₺M / ≈ \$M)	134 (~\$2.82M)	260 (~\$5.47M)	354 (~\$7.45M)
Unit cash cost (₺ / ≈ \$)	~311,000 (~\$6,547)	~307,000 (~\$6,463)	~297,000 (~\$6,253)
Unit EBITDA (₺ / ≈ \$)	~189,000 (~\$3,979)	~213,000 (~\$4,484)	~248,000 (~\$5,221)
EBITDA (₺M / ≈ \$M)	50.7 (~\$1.07M)	106.5 (~\$2.24M)	161.2 (~\$3.39M)
EBITDA margin	38%	41%	46%

Investment and free cash flow (₺M ≈ \$M)

	Year 1	Year 2	Year 3
EBITDA	50.7 (~\$1.07M)	106.5 (~\$2.24M)	161.2 (~\$3.39M)
Corporate tax (25%, after depreciation)	– 12 (~\$253K)	– 25 (~\$526K)	– 38 (~\$800K)
Capital expenditure	– 33.5 (~\$705K)	– 40 (~\$842K)	– 11 (~\$232K)

	Year 1	Year 2	Year 3
Free cash flow	+5.2 (~\$109K)	+41.5 (~\$874K)	+112.2 (~\$2.36M)
Cumulative free cash flow	+5.2 (~\$109K)	+46.7 (~\$983K)	+158.9 (~\$3.35M)

Results:

- **Simple payback:** ~15 months (from production start)
- **3-year cumulative free cash flow:** ~£159M (~\$3.35M) — ~1.9× the investment
- **Project IRR (discounted):** ~40%+ (≥20% in downside scenario, see Section 8)

The entry margin is ~38% (due to the competitive entry price); it rises to ~46% by Y3 through scale, price normalization and automation. The margin rests on: (a) low steel input, high value-added product, (b) customs protection against imports, (c) low-cost skilled labor plus the external cutting model. In a "price -15%" scenario the margin falls to ~27% (see sensitivity).

8. SENSITIVITY ANALYSIS

Scenario	EBITDA margin	Payback estimate
Base	~38% (entry) → 46% (Y3)	~15 months
Price -15%	~27%	~23 months
Volume -25%	~33%	~19 months
Price -15% and volume -25%	~23%	~29 months

The model is most sensitive to **price** (steel is not the top lever because the steel share of unit cost is low relative to the price share). If steel rises +20%, unit cost rises from ~£317,000 (~\$6,674) to ~£343,000 (~\$7,221); the margin can be protected via price adjustment.

9. STAFFING PLAN

Phase	Total	Direct (welding/assembly/paint)	Indirect (engineering/QC/sales/admin)
Phase 1	15	9	6
Phase 2	20	12	8
Phase 3	19	10	9

Phase 1 salary structure (steady 40 units/month — fully loaded employer cost)

Position	Staff	Gross salary (£/month)	Fully loaded cost (£/month)	Subtotal (£/month)
Direct (9)				
Welder (MIG/MAG)	4	~60,500	~76,000	304,000
Fitter / assembler	3	~60,500	~76,000	228,000

Position	Staff	Gross salary (£/month)	Fully loaded cost (£/month)	Subtotal (£/month)
Painter (2K)	1	~43,800	~55,000	55,000
Crane / assistant operator	1	~48,500	~61,000	61,000
Direct subtotal	9	—	—	648,000
Indirect (6)				
Quality control	1	~58,900	~74,000	74,000
Drafting / CAD	1	~58,900	~74,000	74,000
Sales / marketing	1	~58,900	~74,000	74,000
Procurement / materials	1	~57,300	~72,000	72,000
Warehouse / logistics	1	~55,700	~70,000	70,000
Admin / accounting	1	~54,100	~68,000	68,000
Indirect subtotal	6	—	—	432,000
Management / technical staff (outside the 15)				
Production / plant manager	1	~121,000	~152,000	152,000
Quality / technical office manager	1	~101,100	~127,000	127,000
Process / machine engineer	1	~95,500	~120,000	120,000
Project / planning coordinator	1	~80,400	~101,000	101,000
Management subtotal	4	—	—	500,000

Fully loaded cost = gross + employer SSI 17.5% + employer unemployment 2% + severance provision 8.3% $\approx$ gross $\times$ ~1.26. This table feeds the Phase 1 unit cost lines **Direct labor £16,200/unit** (648,000 $\div$ 40), **Indirect labor £10,800/unit** (432,000 $\div$ 40) and **Management/technical staff £12,500/unit** (500,000 $\div$ 40). **Note — start-up phase pay:** Management/technical salaries are deliberately kept conservative on the premise that the founder-operator of a newly established plant works below market during the ramp-up and reinvests the difference; as volume and profitability mature (Phase 2–3) pay is normalized to market.

In Phase 2, direct staff rises from 9 to 12 (welders 4→6, +2 painters/operators) and indirect from 6 to 8. In Phase 3, welders drop from 6 to 2; a robot operator and a welding engineer are added → 19 total.

Welding starts with **4 welders** in Phase 1 — building a customer portfolio realistically takes ~1 year, so initial production is 15–20 units/month (no headcount kept for 40 units/month). Welding grows to 6 in Phase 2 with volume. In Phase 3 robots take over most welding work and welders drop to **2**; a single **robot operator** and a welding engineer are added → robots lower headcount while capacity rises to 55–60 units/month.

10. IMPLEMENTATION TIMELINE

Month	Step
1–2	Company setup, facility lease/fit-out, crane installation, equipment orders, first hires
3	Portable cutting + fixtures commissioned, first prototype, customer approval

Month	Step
4–6	Serial production start (15–20 units/month), sales network, quality system
7–12	Ramp (15–36 units/month) · Gate 1 (sales ≥ 30 units/month)
13–24	Phase 2: CNC plasma commissioned, capacity 45 units/month · Gate 2 (sales ≥ 45 units/month)
25–30	Phase 3: robot + positioner installation + commissioning
30–36	Full capacity 55–60 units/month + export launch (Europe / Middle East / North Africa)

11. RISKS AND MITIGATIONS

Risk	Likelihood	Impact	Mitigation
Demand below expectation	Medium	High	Phased investment; go/no-go gate at every phase
Chinese import competition	High	Medium	Customs protection; positioning on short lead time + warranty + customization
Steel price increase	Medium	Medium	Long-term supply contracts, scrap hedging, index-linked sales
FX volatility	Medium	Low	Inputs fully domestic; export sales in EUR/USD
Skilled labor shortage	Medium	Medium	In-house training + robotization (Phase 3)
Working capital pressure	Medium	Medium	Production synchronized to demand, receivables management, factoring
Quality / scrap	Low	Medium	Fixture-based assembly, measuring/QC procedures, robot welding

12. INVESTOR VALUE PROPOSITION

- Low-risk start:** Production begins with the first ₺30M (~\$632K); the full ₺84.5M (~\$1.78M) is not committed before demand is proven.
- Self-financed scaling:** Later phases are financed from the current phase's cash flow; external debt needs are low.
- Protected market:** Local production operates under customs protection vs imports; "domestic product" status in public tenders.
- Geographic advantage:** 2–5 days freight to Europe / Middle East / North Africa exports (30+ days from China).
- Competitive entry, short payback:** ~38% entry margin (Y1) → ~46% normalization margin (Y3); ~15-month simple payback.
- Clear exit options:** Stop/downsize option at every phase; limited stranded-asset risk.
- Price penetration:** Enter ~13% below domestic competitors; growth comes from volume, price normalizes in Y2/Y3.

13. DISCLAIMER

All figures in this document are **estimates** based on publicly available market data as of August 2026 (minimum wage, steel prices, FX rates, tax rates). Actual results may differ depending on demand, steel and FX prices, competition and operational performance. This document is not investment advice; independent audit and legal review are recommended before any decision.